ASSIGNMENT-20.2

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BATCH-30

PROMPT: Generate a basic Python program (no external libraries) that demonstrates multiple cybersecurity improvements in a single menudriven application.

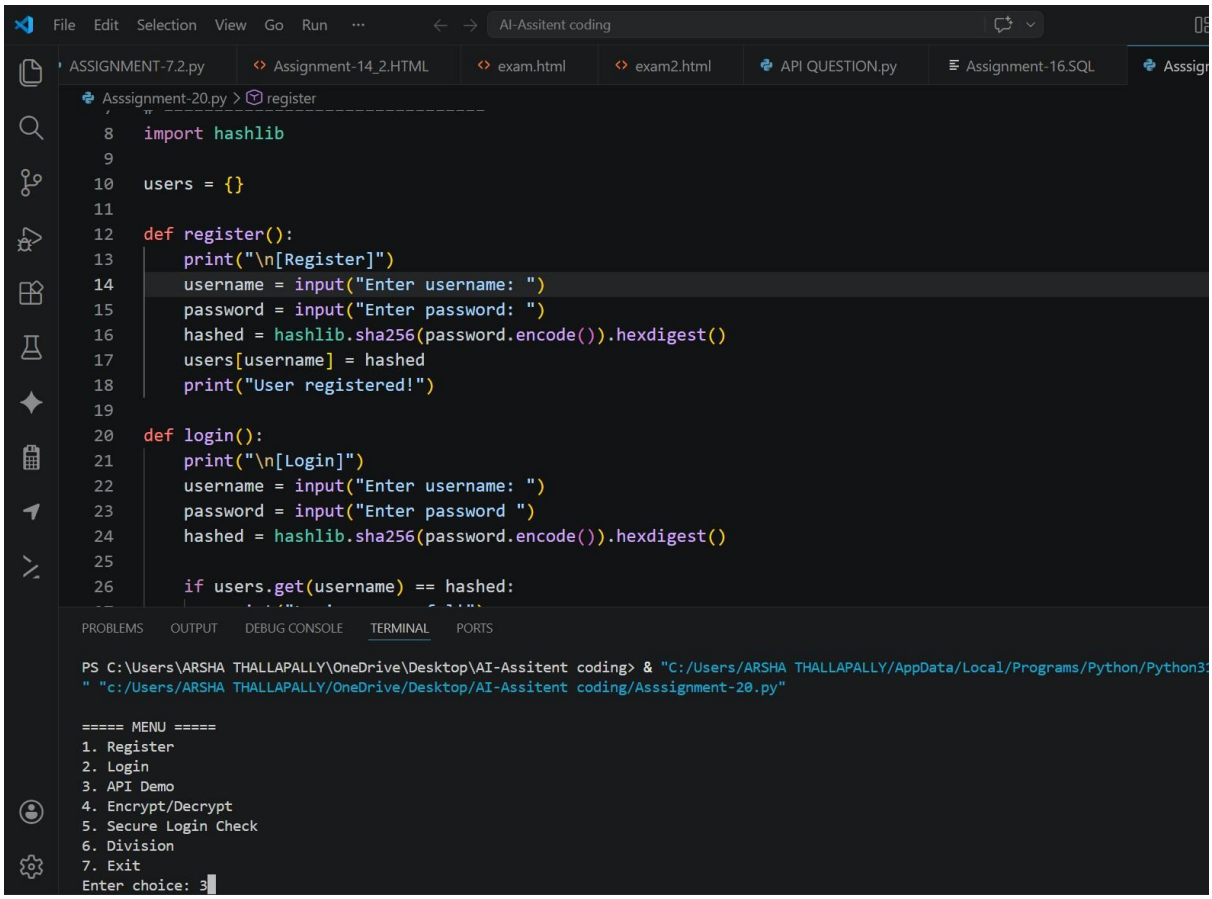
The program should include the following functionalities:

1. Password Security:

- Allow user registration and login

- Store passwords securely using hashing (hashlib with SHA-256)

- Do not store plain text passwords CODE:

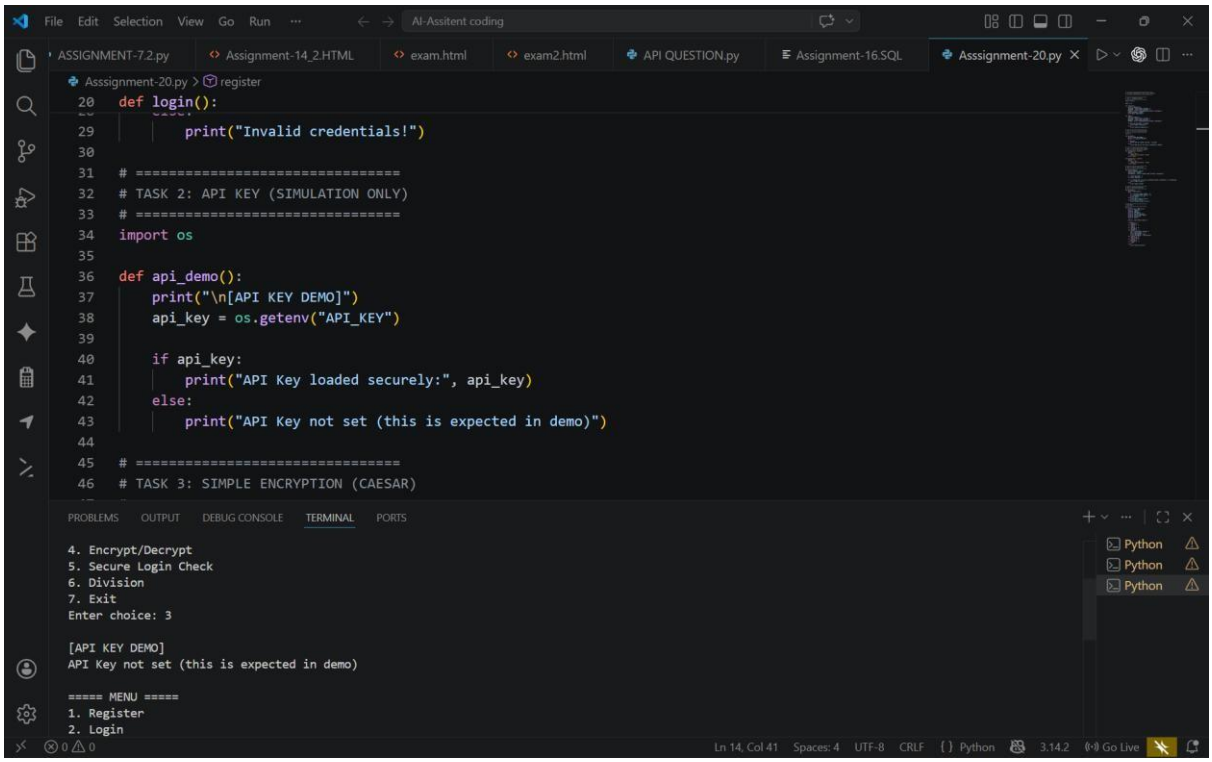


PROMPT:

2. Secure API Key Handling:

- Demonstrate fetching an API key using environment variables (os module)

- If not set, display a message instead of crashing CODE:



PROMPT:

3. Encryption and Decryption:

- Implement a simple encryption and decryption method (Caesar cipher)

- Allow user to input a message and display encrypted and decrypted output

```python
def login():
        print("Invalid credentials!")

# ================================
# TASK 2: API KEY (SIMULATION ONLY)
# ================================
import os

def api_demo():
    print("\n[API KEY DEMO]")
    api_key = os.getenv("API_KEY")

    if api_key:
        print("API Key loaded securely:", api_key)
    else:
        print("API Key not set (this is expected in demo)")

# ================================
# TASK 3: SIMPLE ENCRYPTION (CAESAR)
```

```
1. Register
2. Login
3. API Demo
4. Encrypt/Decrypt
5. Secure Login Check
6. Division
7. Exit
Enter choice: 4
Enter message: MJHF
Encrypted: PMKI
Decrypted: MJHF
```

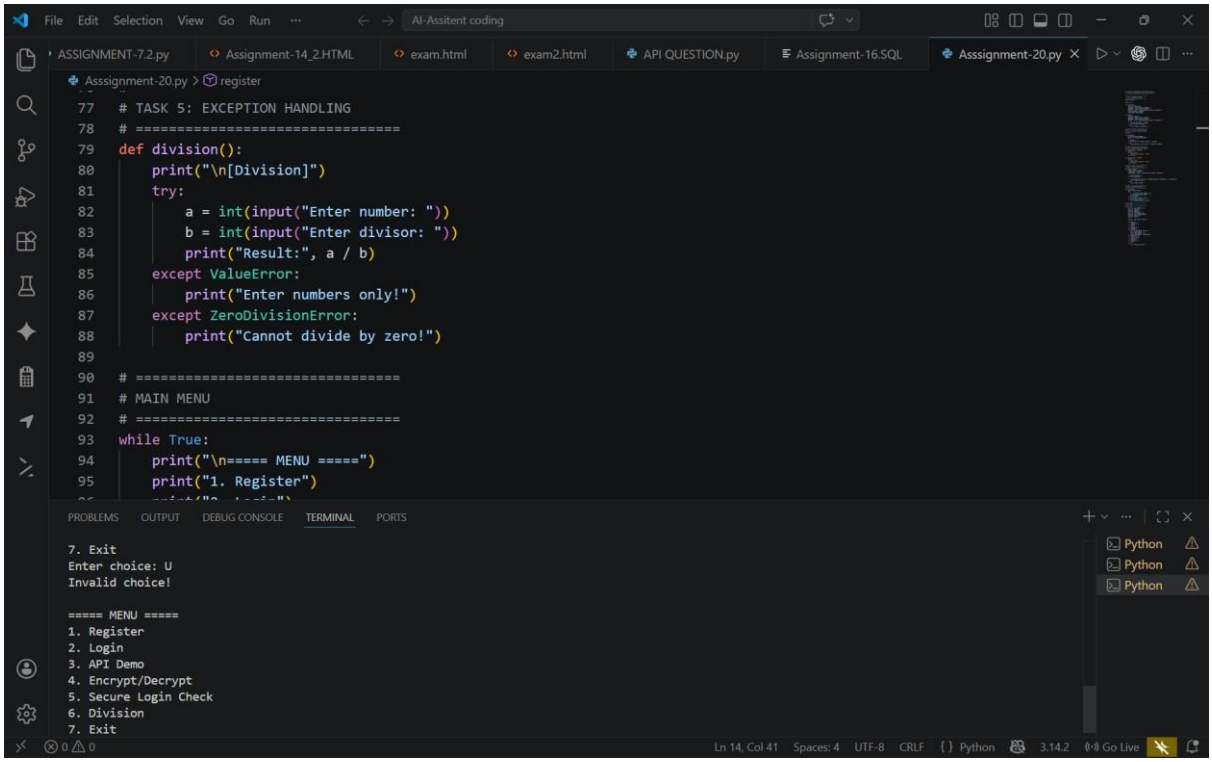
PROMPT:4. Secure Login Check:- Simulate a login system with a predefined username and hashed password

```python
# ================================
# TASK 4: SIMPLE LOGIN CHECK
# ================================
def secure_login():
    print("\n[Secure Login]")
    stored_user = "admin"
    stored_pass = hashlib.sha256("1234".encode()).hexdigest()

    u = input("Username: ")
    p = input("Password: ")

    if u == stored_user and hashlib.sha256(p.encode()).hexdigest() == stored_pass:
        print("Login success")
    else:
        print("Login failed")

# ================================
# TASK 5: EXCEPTION HANDLING
# ================================
def division():
```

```
4. Encrypt/Decrypt
5. Secure Login Check
6. Division
7. Exit
Enter choice: U
Invalid choice!

===== MENU =====
1. Register
2. Login
3. API Demo
4. Encrypt/Decrypt
```

- Validate user input securely

PROMPT:5. Exception Handling:

- Take user input for division

- Handle errors like invalid input and division by zero using try-except